Solutions to Linear Algebra Problems

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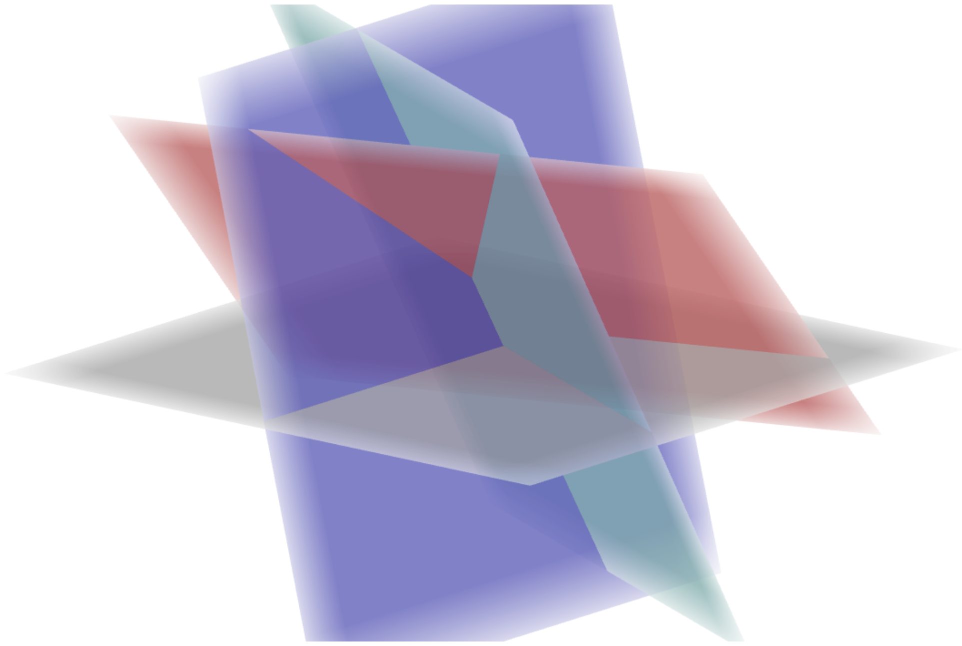
Abstract

This note collects my solutions to selected problems in Linear Algebra. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Linear Algebra Done Right 4e, Sheldon Axler



Problem 1.

Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Source: Linear Algebra Done Right - Ex.12 p.25

Solution:

Let V_1, V_2 be two subspaces of V.
($\Rightarrow$): Suppose that $V_1 \cup V_2$ is a subspace of V. Suppose for contradiction that one of the subspaces V_1 or V_2 is not contained in the other. We can choose u, w such that $u \in V_1, u \notin V_2$ and $w \in V_2, w \notin V_1$.
Since $V_1 \cup V_2$ is a subspace, we have $u + w \in V_1 \cup V_2$. This means $u + w$ must be in V_1 or V_2.
If $u + w \in V_1$, since V_1 is a subspace and $-u \in V_1$, then $w = (u + w) - u \in V_1$, which is a contradiction. Similarly, if $u+w \in V_2$, since V_2 is a subspace and $-w \in V_2$, then $u = (u+w)-w \in V_2$, which is also a contradiction.
Therefore, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.
($\Leftarrow$): If $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$, then $V_1 \cup V_2 = V_2$ or $V_1 \cup V_2 = V_1$, which is a subspace.

Problem 2.

Suppose $v_1, \ldots, v_m$ is a list of vectors in V. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k.$$

Show that the list $v_1, \ldots, v_m$ is linearly independent if and only if the list $w_1, \ldots, w_m$ is linearly independent.

Source: Linear Algebra Done Right - Ex.14 p.38

Solution:

($\Rightarrow$): Suppose the list $v_1, \ldots, v_m$ is linearly independent. Consider a combination of $w_1, \ldots, w_m$ such that:

$$\begin{aligned} c_1w_1 + c_2w_2 + \cdots + c_mw_m &= 0 \\ \Rightarrow c_1v_1 + c_2(v_1 + v_2) + \cdots + c_m(v_1 + v_2 + \cdots + v_m) &= 0 \\ \Rightarrow (c_1 + c_2 + \cdots + c_m)v_1 + (c_2 + \cdots + c_m)v_2 + \cdots + c_mv_m &= 0 \end{aligned}$$

Since $v_1, \ldots, v_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $w_1, \ldots, w_m$ are linearly independent.
($\Leftarrow$): Suppose $w_1, \ldots, w_m$ are linearly independent. Consider a combination of $v_1, \ldots, v_m$ such

that:

$$\begin{aligned}c_1v_1 + c_2v_2 + \cdots + c_mv_m &= 0\\ \Rightarrow c_1w_1 + c_2(w_2 - w_1) + c_3(w_3 - w_2) + \cdots + c_m(w_m - w_{m-1}) &= 0\\ \Rightarrow (c_1 - c_2)w_1 + (c_2 - c_3)w_2 + \cdots + (c_{m-1} - c_m)w_{m-1} + c_mw_m &= 0\end{aligned}$$

Since $w_1, \dots, w_m$ are linearly independent, it follows that $c_1 = c_2 = \cdots = c_m = 0$. Therefore, the list $v_1, \dots, v_m$ are linearly independent.

Problem 3.

Suppose V is finite-dimensional and U, W are subspaces of V such that $V = U + W$. Prove that there exists a basis of V consisting of vectors in $U \cup W$.

Source: Linear Algebra Done Right - Ex.5 p.43

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U and $w_1, w_2, \dots, w_n$ be a basis of W.
Since $V = U + W$, we have the list $\{u_1, u_2, \dots, u_m, w_1, w_2, \dots, w_n\}$ spans V.
Applying the procedure in Theorem 2.30 to reduce this list to a basis of V produces a basis consisting of the vectors $u_1, \dots, u_m$ and some of the w's (none of the u's get deleted in this procedure because $u_1, \dots, u_m$ is linearly independent).

Problem 4.

Suppose V is finite-dimensional and U is a subspace of V with $U \neq V$. Let $n = \dim V$ and $m = \dim U$. Prove that there exist $n - m$ subspaces of V, each of dimension $n - 1$, whose intersection equals U.

Source: Linear Algebra Done Right - Ex.16 p.49

Solution:

Let $u_1, u_2, \dots, u_m$ be a basis of U. This list can be extended to a basis of V by adjoining $n - m$ additional vectors (by Theorem 2.32), say $w_1, w_2, \dots, w_{n-m}$. We denote this extended list as L.
Let V_i $(1 \leq i \leq n-m)$ be the subspace spanned by L without including w_i. Clearly $\dim V_i = n-1$ and $\bigcap_{i=1}^{n-m} V_i$ is also a subspace that contains U.

Consider $v \in \bigcap_{i=1}^{n-m} V_i$. We have:

$$a_1u_1 + a_2u_2 + \dots + a_mu_m + b_2w_2 + b_3w_3 + \dots$$
$$+ b_{n-m}w_{n-m} = v \quad (\text{since } v \in V_1)$$

$$c_1u_1 + c_2u_2 + \dots + c_mu_m + d_1w_1 + d_3w_3 + d_4w_4$$
$$+ \dots + d_{n-m}w_{n-m} = v \quad (\text{since } v \in V_2)$$

By subtracting, we have:

$$(a_1 - c_1)u_1 + (a_2 - c_2)u_2 + \dots + (a_m - c_m)u_m - d_1w_1$$
$$+ (b_2 - d_2)w_2 + b_3w_3 + \dots + (b_{n-m} - d_{n-m})w_{n-m} = 0$$

Since L is a list of independent vectors and by repeating this process, this implies that all coefficients for w_i's are 0 and v is uniquely represented in the basis of U.
We conclude that $\bigcap_{i=1}^{n-m} V_i = U$.

Problem 5.

Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V. In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that $Tu = Su$ for all $u \in U$.

Source: Linear Algebra Done Right - Ex.13 p.58

Solution:

Let U be a subspace of V, let $S \in \mathcal{L}(U, W)$, and let $u_1, u_2, \dots, u_n$ be a basis of U. By Theorem 2.32, this list can be extended to a basis of V:

$$u_1, u_2, \dots, u_n, v_{n+1}, v_{n+2}, \dots, v_m.$$

For $1 \leq i \leq n$, let

$$w_i = Su_i.$$

Choose arbitrary vectors $w_{n+1}, w_{n+2}, \dots, w_m \in W$. By the linear map lemma, there exists a unique $T \in \mathcal{L}(V, W)$ such that

$$Tu_1 = w_1 = Su_1, \quad \dots, \quad Tu_n = w_n = Su_n,$$

and

$$Tv_{n+1} = w_{n+1}, \quad \ldots, \quad Tv_m = w_m.$$

Now let $u \in U$. Since $u_1, \ldots, u_n$ is a basis of U, there exist scalars $a_1, \ldots, a_n$ such that

$$u = a_1u_1 + a_2u_2 + \cdots + a_nu_n.$$

Thus

$$\begin{aligned} Tu &= T(a_1u_1 + a_2u_2 + \cdots + a_nu_n) \\ &= a_1Tu_1 + a_2Tu_2 + \cdots + a_nTu_n \\ &= a_1Su_1 + a_2Su_2 + \cdots + a_nSu_n \\ &= S(a_1u_1 + a_2u_2 + \cdots + a_nu_n) \\ &= Su. \end{aligned}$$

Therefore, $Tu = Su$ for all $u \in U$, so T is an extension of S to V.

Problem 6.

Suppose V and W are finite-dimensional and that U is a subspace of V. Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U$ if and only if $\dim U \geq \dim V - \dim W$.

Source: Linear Algebra Done Right - Ex.18 p.67

Solution:

($\Rightarrow$) Suppose there exists $T \in \mathcal{L}(V, W)$ such that null $T = U$. By the fundamental theorem of linear maps, we have:

$$\dim V = \dim \text{null } T + \dim \text{range } T$$

$$\Rightarrow \dim U = \dim V - \dim \text{range } T \geq \dim V - \dim W$$

($\Leftarrow$) Suppose U is a subspace of V such that $\dim U \geq \dim V - \dim W$. Let $u_1, u_2, \ldots, u_n$ be a basis of U. This list can be extended to $u_1, u_2, \ldots, u_n, v_1, v_2, \ldots, v_m$ to be a basis of $V \Rightarrow \dim V = m+n$. We have $\dim U \geq \dim V - \dim W \Rightarrow \dim W \geq \dim V - \dim U = m$. Therefore, we can let $w_1, w_2, \ldots, w_m$ be an independent list of vectors in W. Define $T : V \to W$ as:

$$T(c_1u_1 + \cdots + c_nu_n + d_1v_1 + \cdots + d_mv_m) = d_1w_1 + d_2w_2 + \cdots + d_mw_m$$

It is easy to verify that T is a linear map by checking $T(u+v) = T(u)+T(v)$ and $T(\lambda u) = \lambda T(u)$, $\forall u, v \in V$.

We have for all $u \in U$, $Tu = T(c_1u_1 + \cdots + c_nu_n) = 0w_1 + \cdots + 0w_m = 0 \Rightarrow u \in \text{null } T \Rightarrow U \subseteq \text{null } T$.

Now, let $u \in \text{null } T$. We have $Tu = 0 \Rightarrow d_1 = d_2 = \cdots = d_m = 0$ since $w_1, w_2, \ldots, w_m$ are independent. Thus, u can be expressed as $c_1u_1 + c_2u_2 + \cdots + c_nu_n \Rightarrow u \in U \Rightarrow \text{null } T \subseteq U$. Therefore, $U = \text{null } T$.

Problem 7.

Suppose there exists a linear map on V whose null space and range are both finite-dimensional. Prove that V is finite-dimensional.

Source: Source: Linear Algebra Done Right - Ex.15 p.66

Solution:

Let $u_1, u_2, \ldots, u_m$ be a basis of null T and $v_1, v_2, \ldots, v_n$ be a basis of range T. Let $w_1, w_2, \ldots, w_n \in V$ such that $v_1 = Tw_1, v_2 = Tw_2, \ldots, v_n = Tw_n$.
For any $z \in V$, we have:

$$\begin{aligned} Tz &= a_1Tw_1 + a_2Tw_2 + \cdots + a_nTw_n \\ &= T(a_1w_1 + a_2w_2 + \cdots + a_nw_n) \end{aligned}$$

Let $u = z - a_1w_1 - a_2w_2 - \cdots - a_nw_n \implies Tu = 0 \implies u \in \text{null } T$
$\implies u = b_1u_1 + b_2u_2 + \cdots + b_mu_m$
$\implies z = a_1w_1 + a_2w_2 + \cdots + a_nw_n + b_1u_1 + b_2u_2 + \cdots + b_mu_m$
This implies that the list $u_1, u_2, \ldots, u_m, w_1, w_2, \ldots, w_n$ spans V.
Therefore, V is finite dimensional.